

# BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

Name:

ID:

**Due Date**

Thursday, 12/04/24, 9:00pm

## Overview

### Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

### Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `c:\Users\Donny\OneDrive\Post-Grad\Courses\BEE 5750\Homework 5\hw5-d
Installed Measures          v0.3.3
Installed GR_jll            v0.73.18+0
Installed OpenBLAS32_jll    v0.3.29+0
```

```

Installed PlotUtils          v1.4.4
Installed OpenSSL            v1.6.0
Installed StaticArrays       v1.9.15
Installed MutableArithmetics v1.6.7
Installed HiGHS_jll          v1.12.0+0
Installed FFMPEG             v0.4.5
Installed StaticArraysCore   v1.4.4
Installed Pango_jll          v1.57.0+0
Installed JSON               v1.3.0
Installed METIS_jll          v5.1.3+0
Installed GR                 v0.73.18
Installed JuMP               v1.29.3
Installed DataStructures     v0.19.3
Installed Graphs             v1.13.1
Installed Interpolations     v0.16.2
Installed Adapt             v4.4.0
Installed StableRNGs         v1.0.4
Installed GraphRecipes       v0.5.15
Installed ChainRulesCore     v1.26.0
Installed FFMPEG_jll         v8.0.0+0
Installed SimpleTraits       v0.9.5
Installed StatsBase          v0.34.8
Installed StructUtils        v2.6.0
Installed HiGHS              v1.20.1
Installed ForwardDiff        v1.3.0
Precompiling project...
 8750.7 ms Measures
 4744.3 ms Adapt
 4742.1 ms StaticArraysCore
 4775.5 ms StableRNGs
 4685.8 ms METIS_jll
 4725.4 ms StructUtils
 4712.5 ms SimpleTraits
 4716.5 ms OpenBLAS32_jll
 4723.3 ms EarCut_jll
 5043.9 ms ChainRulesCore
 5080.3 ms DataStructures
1181.0 ms Adapt → AdaptSparseArraysExt
1098.3 ms StructUtils → StructUtilsTablesExt
1176.2 ms OffsetArrays → OffsetArraysAdaptExt
1171.8 ms DiffResults
 6924.9 ms MutableArithmetics
1467.3 ms ColorVectorSpace → SpecialFunctionsExt

```

|            |                                                      |
|------------|------------------------------------------------------|
| 748.7 ms   | SortingAlgorithms                                    |
| 2136.3 ms  | HiGHS_jll                                            |
| 2137.0 ms  | ChainRulesCore → ChainRulesCoreSparseArraysExt       |
| 2374.9 ms  | LogExpFunctions → LogExpFunctionsChainRulesCoreExt   |
| 3739.2 ms  | OpenSSL                                              |
| 2377.7 ms  | SpecialFunctions → SpecialFunctionsChainRulesCoreExt |
| 4957.2 ms  | Pango_jll                                            |
| 5104.9 ms  | JSON                                                 |
| 5344.8 ms  | ColorSchemes                                         |
| 3842.2 ms  | ForwardDiff                                          |
| 3494.0 ms  | StatsBase                                            |
| 5628.3 ms  | FFMPEG_jll                                           |
| 937.6 ms   | FFMPEG                                               |
| 1779.6 ms  | BenchmarkTools                                       |
| 12550.5 ms | GR_jll                                               |
| 12728.7 ms | PlotUtils                                            |
| 18171.6 ms | StaticArrays                                         |
| 15246.9 ms | HTTP                                                 |
| 1391.5 ms  | Adapt → AdaptStaticArraysExt                         |
| 1422.1 ms  | StaticArrays → StaticArraysStatisticsExt             |
| 2042.0 ms  | StaticArrays → StaticArraysChainRulesCoreExt         |
| 1784.2 ms  | ForwardDiff → ForwardDiffStaticArraysExt             |
| 2199.5 ms  | ArnoldiMethod                                        |
| 3377.8 ms  | PlotThemes                                           |
| 4552.0 ms  | RecipesPipeline                                      |
| 2755.5 ms  | GeometryTypes                                        |
| 2918.7 ms  | Interpolations                                       |
| 4623.0 ms  | GR                                                   |
| 1521.0 ms  | Interpolations → InterpolationsForwardDiffExt        |
| 4783.4 ms  | Graphs                                               |
| 11197.8 ms | GeometryBasics                                       |
| 1741.2 ms  | NetworkLayout                                        |
| 2491.8 ms  | NetworkLayout → NetworkLayoutGraphsExt               |
| 36076.8 ms | DataFrames                                           |
| 2340.3 ms  | Latexify → DataFramesExt                             |
| 6843.3 ms  | GraphRecipes                                         |
| 36656.4 ms | MathOptInterface                                     |
| 2079.8 ms  | MathOptIIS                                           |
| 9228.4 ms  | HiGHS                                                |
| 15737.2 ms | JuMP                                                 |
| 54685.0 ms | Plots                                                |
| 3917.8 ms  | Plots → GeometryBasicsExt                            |

59 dependencies successfully precompiled in 115 seconds. 193 already precompiled.

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables

```

## Problems (Total: 30 Points)

### Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

| City/Facility | Landfill (km) | MRF (km) | WTE (km) |
|---------------|---------------|----------|----------|
| 1             | 5             | 30       | 15       |
| 2             | 15            | 25       | 10       |
| 3             | 13            | 45       | 20       |
| LF            | -             | 32       | 18       |
| MRF           | 32            | -        | 15       |
| WTE           | 18            | 15       | -        |

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

| Component   | % of total mass | Combustion ash (%) | MRF Recycling rate (%) |
|-------------|-----------------|--------------------|------------------------|
| Food Wastes | 15              | 8                  | 0                      |

| Component         | % of total mass | Combustion ash (%) | MRF Recycling rate (%) |
|-------------------|-----------------|--------------------|------------------------|
| Paper & Cardboard | 40              | 7                  | 55                     |
| Plastics          | 5               | 5                  | 15                     |
| Textiles          | 3               | 10                 | 10                     |
| Rubber, Leather   | 2               | 15                 | 0                      |
| Wood              | 5               | 2                  | 30                     |
| Yard Wastes       | 18              | 2                  | 40                     |
| Glass             | 4               | 100                | 60                     |
| Ferrous           | 2               | 100                | 75                     |
| Aluminum          | 2               | 100                | 80                     |
| Other Metal       | 1               | 100                | 50                     |
| Miscellaneous     | 3               | 70                 | 0                      |

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

**Reminder:** Use `round(x; digits=n)` to report values to the appropriate precision!

### Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

### Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

### Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

### Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

### Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

```
# Initialize the Model
model = Model(HiGHS.Optimizer)

# SETS & PARAMETERS
# Cities (1, 2, 3) and Facilities (LF, MRF, WTE)
cities = 1:3
facilities = ["LF", "MRF", "WTE"]

# Waste Generation (Mg/day)
gen = [100, 90, 120]

# Capacities (Mg/day)
cap = Dict{"LF" => 200, "MRF" => 350, "WTE" => 210}

# Fixed Costs ($/day)
fixed_cost = Dict{"LF" => 2000, "MRF" => 1500, "WTE" => 2500}

# Tipping Costs ($/Mg)
tipping_cost = Dict{"LF" => 50, "MRF" => 7, "WTE" => 60}

# Recycling Cost ($/Mg recycled)
recycle_proc_cost = 40

# Process Parameters
r_overall = 0.3775      # Recycling rate
r_residual = 0.6225     # Residual rate (1 - r)
ash_raw = 0.1641        # Ash from raw waste
ash_res = 0.1386        # Ash from MRF residuals

# Transport Cost ($/Mg-km)
trans_rate = 1.5

# Distances (City to Facility)
# Rows: Cities 1, 2, 3. Cols: LF, MRF, WTE
dist_city_fac = [
    5 30 15;
    15 25 10;
    13 45 20
```

```

]

# Distances (Between Facilities)
dist_mrf_lf = 32
dist_mrf_wte = 15
dist_wte_lf = 18

# DECISION VARIABLES
# x[i, j]: Waste from City i to Facility j
@variable(model, x[1:3, 1:3] >= 0)

# y_flow: Specific facility-to-facility flows
@variable(model, y_mrf_lf >= 0)
@variable(model, y_mrf_wte >= 0)
@variable(model, y_wte_lf >= 0)

# z[j]: Binary status (1 if open, 0 if closed)
@variable(model, z[1:3], Bin)

# OBJECTIVE FUNCTION
# 1. Transport Costs
transport_cost = sum(trans_rate * dist_city_fac[i,j] * x[i,j] for i in 1:3, j in 1:3) +
    trans_rate * dist_mrf_lf * y_mrf_lf +
    trans_rate * dist_mrf_wte * y_mrf_wte +
    trans_rate * dist_wte_lf * y_wte_lf

# 2. Fixed Costs (Indices: 1=LF, 2=MRF, 3=WTE)
fac_fixed_cost = fixed_cost["LF"]*z[1] + fixed_cost["MRF"]*z[2] + fixed_cost["WTE"]*z[3]

# 3. Variable/Processing Costs
# LF Cost (Inputs: Direct x + MRF resid + WTE ash)
cost_lf = tipping_cost["LF"] * (sum(x[i,1] for i in 1:3) + y_mrf_lf + y_wte_lf)

# MRF Cost (Inputs: Direct x) + Recycling Processing Cost
# Note: Recycling cost applies to the recycled fraction (r_overall * input)
cost_mrf = tipping_cost["MRF"] * sum(x[i,2] for i in 1:3) +
    recycle_proc_cost * (r_overall * sum(x[i,2] for i in 1:3))

# WTE Cost (Inputs: Direct x + MRF resid)
cost_wte = tipping_cost["WTE"] * (sum(x[i,3] for i in 1:3) + y_mrf_wte)

@objective(model, Min, transport_cost + fac_fixed_cost + cost_lf + cost_mrf + cost_wte)

```

```

# CONSTRAINTS

# 1. Waste Generation (Supply)
for i in 1:3
    @constraint(model, sum(x[i,j] for j in 1:3) == gen[i])
end

# 2. Mass Balance: MRF Residuals
# Output residuals = Input * (1 - recycling_rate)
@constraint(model, y_mrf_lf + y_mrf_wte == r_residual * sum(x[i,2] for i in 1:3))

# 3. Mass Balance: WTE Ash
# Ash = (Raw Input * Raw Ash Rate) + (Residual Input * Residual Ash Rate)
@constraint(model, y_wte_lf == ash_raw * sum(x[i,3] for i in 1:3) + ash_res * y_mrf_wte)

# 4. Capacity & Linking Constraints (Big M)
# LF (Index 1)
@constraint(model, sum(x[i,1] for i in 1:3) + y_mrf_lf + y_wte_lf <= cap["LF"] * z[1])

# MRF (Index 2)
@constraint(model, sum(x[i,2] for i in 1:3) <= cap["MRF"] * z[2])

# WTE (Index 3)
@constraint(model, sum(x[i,3] for i in 1:3) + y_mrf_wte <= cap["WTE"] * z[3])

# SOLVE
optimize!(model)

# REPORT
println("Status: ", termination_status(model))
println("Optimal Objective Cost: \$", objective_value(model))

println("\nFacility Status:")
println("LF Open? ", value(z[1]))
println("MRF Open? ", value(z[2]))
println("WTE Open? ", value(z[3]))

println("\nPrimary Flows (City -> Facility):")
for i in 1:3
    println("City $i -> LF: ", value(x[i,1]))
    println("City $i -> MRF: ", value(x[i,2]))
    println("City $i -> WTE: ", value(x[i,3]))
end

```

```

end

println("\nSecondary Flows:")
println("MRF -> LF: ", value(y_mrf_lf))
println("MRF -> WTE: ", value(y_mrf_wte))
println("WTE -> LF: ", value(y_wte_lf))

```

Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT licence terms  
MIP has 8 rows; 15 cols; 34 nonzeros; 3 integer variables (3 binary)

Coefficient ranges:

```

Matrix [1e-01, 4e+02]
Cost    [6e+01, 2e+03]
Bound   [1e+00, 1e+00]
RHS     [9e+01, 1e+02]

```

Presolving model

8 rows, 15 cols, 34 nonzeros 0s

7 rows, 13 cols, 31 nonzeros 0s

Presolve reductions: rows 7(-1); columns 13(-2); nonzeros 31(-3)

Solving MIP model with:

7 rows

13 cols (2 binary, 0 integer, 0 implied int., 11 continuous, 0 domain fixed)

31 nonzeros

Src: B => Branching; C => Central rounding; F => Feasibility pump; H => Heuristic;

I => Shifting; J => Feasibility jump; L => Sub-MIP; P => Empty MIP; R => Randomized round;

S => Solve LP; T => Evaluate node; U => Unbounded; X => User solution; Y => HiGHS solution;

Z => ZI Round; l => Trivial lower; p => Trivial point; u => Trivial upper; z => Trivial upper;

| Nodes |       |         | B&B Tree |       | Objective Bounds |              |        | Dynamic Cuts |          |
|-------|-------|---------|----------|-------|------------------|--------------|--------|--------------|----------|
| Src   | Proc. | InQueue | Leaves   | Expl. | BestBound        | BestSol      | Gap    | Cuts         | Inactive |
| J     | 0     | 0       | 0        | 0.00% | -inf             | 39552.775615 | Large  | 0            |          |
| S     | 0     | 0       | 0        | 0.00% | 2000             | 27855.482115 | 92.82% | 0            |          |

50.0% inactive integer columns, restarting

Model after restart has 1 row, 3 cols (0 bin., 0 int., 0 impl., 3 cont., 0 dom.fix.), and 3 nonzeros

|   |   |   |   |         |              |              |       |   |  |
|---|---|---|---|---------|--------------|--------------|-------|---|--|
| T | 0 | 0 | 0 | 0.00%   | 26922.085548 | 27855.482115 | 3.35% | 0 |  |
|   | 0 | 0 | 0 | 0.00%   | 26922.085548 | 27855.482115 | 3.35% | 0 |  |
|   | 1 | 0 | 1 | 100.00% | 27855.482115 | 27855.482115 | 0.00% | 0 |  |

Solving report

|                   |                           |
|-------------------|---------------------------|
| Status            | Optimal                   |
| Primal bound      | 27855.4821151             |
| Dual bound        | 27855.4821151             |
| Gap               | 0% (tolerance: 0.01%)     |
| P-D integral      | 0.00337140793481          |
| Solution status   | feasible                  |
|                   | 27855.4821151 (objective) |
|                   | 0 (bound viol.)           |
|                   | 0 (int. viol.)            |
|                   | 0 (row viol.)             |
| Timing            | 0.02                      |
| Max sub-MIP depth | 0                         |
| Nodes             | 1                         |
| Repair LPs        | 0                         |
| LP iterations     | 6                         |
|                   | 0 (strong br.)            |
|                   | 0 (separation)            |
|                   | 0 (heuristics)            |

Status: OPTIMAL

Optimal Objective Cost: \$27855.48211508554

Facility Status:

LF Open? 1.0

MRF Open? 0.0

WTE Open? 1.0

Primary Flows (City -> Facility):

City 1 -> LF: 100.0

City 1 -> MRF: 0.0

City 1 -> WTE: 0.0

City 2 -> LF: 0.0

City 2 -> MRF: 0.0

City 2 -> WTE: 90.0

City 3 -> LF: 78.40531164014834

City 3 -> MRF: 0.0

City 3 -> WTE: 41.594688359851666

Secondary Flows:

MRF -> LF: 0.0

MRF -> WTE: 0.0

WTE -> LF: 21.59468835985166

Based on the JuMP solver above, we obtained that the optimal cost based on the Objective Function is **\$27855,48**. This is achieved by 1. City 1 send all of its waste to landfill 2. City 2 send all of its waste to WTE 3. City 3 send 78,4 Mg/day of its waste to landfill and 41,6 Mg/day of its waste to WTE 4. MRF facility will not be operated

### Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

### Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in 'data/generators.csv.' In period 1, the demand is  $d_1 = 1100\text{MW}$ . In period 2, the demand is projected to be  $d_2 = 1200\text{MW}$ , but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

### Problem 2.1

Draw a scenario tree for this problem.

### Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

### References

LLM was incorporated in solving the problems in the following manner

1. Problem 1.5 - After building the decision variable, objective function, and constraints, LLM was incorporated to build the JuMP draft code
2. Problem 2.2 - LLM was used to consult on the constraints for the stochastic formulation